

3 如图4-3-8所示AB梁尺寸(160mm×300mm)及承受载荷情况。试求梁承受的最大应力。

将力分解为法向与轴向两个方向。

$$F_n = F \sin 30^\circ$$

$$F_t = F \cos 30^\circ$$

易知在C处附近,弯矩最大。  $M_c = F_t \cdot l_{AC}$

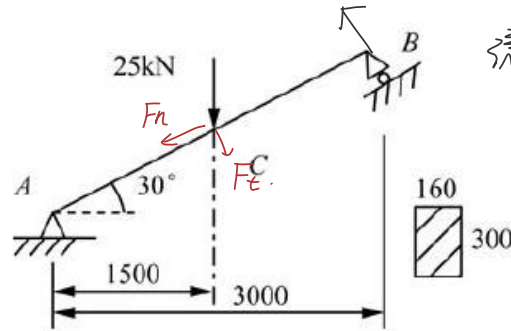
轴力对AC段,轴力均匀  $F_{ACN} = F_n \sin 30^\circ = 12.5 \text{ kN}$ 。

$$\sigma_{AC} = F_{ACN} / A = 0.26 \text{ MPa}$$

$$\sigma_{BC} = F_{BCN} / A = 0.26 \text{ MPa}$$

弯矩: C处边缘:  $\sigma_t = M_c / W_z \approx 7.81 \text{ MPa}$ 。

C上边缘:  $-\sigma_c = M_c / W_z \approx 7.81 \text{ MPa}$ 。



求得

$$\sigma_{ct} = \sigma_c + \sigma_{AC} = -8.07 \text{ MPa}$$

$$\sigma_{ct} = \sigma_t + \sigma_{BC} = 7.55 \text{ MPa}$$

$$\therefore \sigma_{\max} = 8.07 \text{ MPa}$$

4 一工形截面的外伸梁如图4-3-10所示。已知:  $l=600 \text{ mm}$ ,  $a=110 \text{ mm}$ ,  $b=30 \text{ mm}$ ,  $c=80 \text{ mm}$ ,  $F_1=24 \text{ kN}$ ,  $F_2=9 \text{ kN}$ , 材料的许用拉应力  $[\sigma_t]=30 \text{ MPa}$ , 许用压应力  $[\sigma_c]=90 \text{ MPa}$ 。

$$y_2 = \frac{ab \cdot \frac{b}{2} + b(b+c) \cdot \frac{b+c}{2}}{ab + b(b+c)}$$

(1) 若C为工形截面形心, 试求  $y_1$  与  $y_2$  的值;

$$(2) I_z = \frac{1}{12} ab^3 + \frac{1}{12} c b^3 + b \cdot (b \cdot \frac{b}{2} - y_c)^2 + ab \cdot (\frac{b}{2} - y_c)^2 = 5.73 \times 10^6 \text{ mm}^4$$

$$y_1 + y_2 = b + c$$

$$\therefore y_1 = 71.84 \text{ mm}$$

$$y_2 = 38.16 \text{ mm}$$

$$\frac{y_1}{y_2} = 1.88$$

(2) 不计弯曲切应力的影响, 试校核该梁的强度。  $M_A = F_1 \cdot \frac{l}{2} + F_2 \cdot \frac{2}{3} l - F_B \cdot l$   $F_B = 24 \text{ kN}$ 。

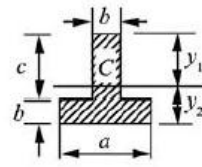
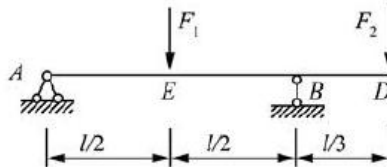
$$F_A = 9 \text{ kN}$$

危险截面有2个:

$$D, E \rightarrow B$$

$$E \text{ 上边缘: } \sigma_c = \frac{M_E y_1}{I_z} \approx 33.35 \text{ MPa} \leq [\sigma_c]$$

$$E \text{ 下边缘: } \sigma_t = \frac{M_E y_2}{I_z} \approx 17.98 \text{ MPa} \leq [\sigma_t]$$



4-34 图4-2-54所示木梁受一可移动的荷载  $F=40 \text{ kN}$  作用。已知许用弯曲正应力  $[\sigma]=10 \text{ MPa}$ , 许用切应力  $[\tau]=3 \text{ MPa}$ 。木梁的横截面为矩形, 其高宽比  $h/b=3/2$ , 试选择梁的截面尺寸。

易知外载作用处弯矩最大。

对某个特定的位置  $x$  处:

$$F = F_A + F_B$$

$$F_x = F_B \cdot l_{AB} \quad F_B = \frac{x}{l_{AB}} F$$

$$F_A = (1 - \frac{x}{l_{AB}}) F$$

$$M(x) = x \cdot F_A = (x - \frac{x^2}{l_{AB}}) F \leq (\frac{l_{AB}}{2} - \frac{l_{AB}^2}{4 l_{AB}}) F = \frac{F l_{AB}}{4} \quad \sigma = \frac{M \cdot \frac{h}{2}}{I_z} = \frac{F l_{AB}}{4 \cdot \frac{b h^3}{6}} \leq [\sigma]$$

$$F_{\text{剪}}(x) = F_A = (1 - \frac{x}{l_{AB}}) F \leq F$$

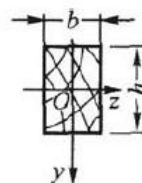
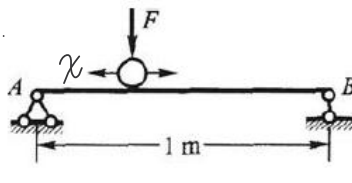
$$\sigma_{\max} = \frac{3}{2} \frac{F_{S, \max}}{A} = \frac{3F}{2bh} \leq [\tau]$$

$$\therefore b \geq 138.67 \text{ mm}$$

$$\therefore b \geq 115.47 \text{ mm}$$

$$\text{综上 } b \geq 138.67 \text{ mm}$$

$$h \geq 208 \text{ mm}$$



$$b \leq c: \sigma_c = \frac{M_E y_1}{I_z} = 22.56 \text{ MPa} \leq [\sigma_c]$$

$$F: \sigma_t = \frac{M_E y_2}{I_z} = 11.98 \text{ MPa} \leq [\sigma_t]$$

$\therefore$  均满足

